

Integer Programming for Large Independent Sets in Sparse Random and Real Networks

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Abstract

We study the maximum independent set problem on sparse Erdős–Rényi graphs $G(n, d/n)$ and on the SNAP ca-GrQc collaboration network. Beyond reporting the size of the independent set returned by an integer program, we pay close attention to the gap between an incumbent solution and a certified optimum. For $d = 20$, random-order greedy tracks the $\log(d)n/d$ scale closely, whereas minimum-degree greedy and IP warm starts recover noticeably larger independent sets. Across the random instances we tested, CBC certifies optimality only at the smallest size, but the incumbent stays comfortably above the greedy scale at every n . On ca-GrQc, decomposing the graph by connected component brings the exact IP within reach and yields $\alpha(G) = 2459$.

1 Problem and Formulation

For a graph $G = (V, E)$, an independent set is a subset $S \subseteq V$ containing no adjacent pair. The independence number is

$$\alpha(G) = \max\{|S| : S \subseteq V, S \text{ independent}\}.$$

The binary IP formulation introduces $x_v \in \{0, 1\}$ for each vertex and solves

$$\begin{aligned} \max \quad & \sum_{v \in V} x_v \\ \text{s.t.} \quad & x_u + x_v \leq 1, \quad (u, v) \in E, \\ & x_v \in \{0, 1\}, \quad v \in V. \end{aligned}$$

These edge constraints are tight in the sense that any feasible binary vector corresponds exactly to an independent set.

In practice we solve the equivalent minimum vertex-cover model. Setting $y_v = 1 - x_v$,

$$\begin{aligned} \min \quad & \sum_{v \in V} y_v \\ \text{s.t.} \quad & y_u + y_v \geq 1, \quad (u, v) \in E, \\ & y_v \in \{0, 1\}, \quad v \in V, \end{aligned}$$

and therefore

$$\alpha(G) = |V| - \tau(G),$$

where $\tau(G)$ is the minimum vertex-cover value. We found CBC noticeably more stable on the cover form once warm starts were enabled. When CBC stops before proving optimality, we report the bracket

$$\alpha(G) \in [L, U],$$

where L is the incumbent independent set size and U is the converted IP upper bound.

2 Random-Graph Benchmarks

The project prompt motivates a classical comparison on sparse random graphs:

$$\alpha(G(n, d/n)) \sim \frac{2 \log d}{d} n$$

in the double limit $n \rightarrow \infty$ followed by $d \rightarrow \infty$, while a range of simple greedy algorithms only reach sets of order $(\log d/d)n$ [2].

2.1 Greedy scale

Random-order greedy gives a clean finite- d heuristic. Process the vertices in a uniformly random order and accept each one whose neighborhood has not already supplied an accepted vertex. Let $s(t)$ be the accepted fraction after a fraction t of the vertices have been exposed. In $G(n, d/n)$, a local mean-field estimate gives

$$\frac{ds}{dt} = \exp(-ds(t)), \quad s(0) = 0,$$

since a fresh vertex is accepted exactly when none of its edges lands in the already-accepted set, which has size roughly $s(t)n$. Integrating yields

$$s(t) = \frac{\log(1 + dt)}{d}, \quad s(1) = \frac{\log(1 + d)}{d}.$$

For $d = 20$ this evaluates to 0.1522, essentially indistinguishable from $\log(20)/20 = 0.1498$. Wormald's differential-equation method is the standard way to make this kind of mean-field approximation rigorous [3].

2.2 First-moment upper benchmark

The asymptotic value $2 \log(d)/d$ is not very accurate numerically at $d = 20$, and a finite- d first-moment bound gives a much sharper reference scale. For $k = an$,

$$\mathbb{E}N_k = \binom{n}{k} (1 - d/n)^{\binom{k}{2}},$$

where N_k is the number of independent sets of size k . Stirling's formula gives

$$\frac{1}{n} \log \mathbb{E}N_{an} = H(a) - \frac{da^2}{2} + o(1),$$

where

$$H(a) = -a \log a - (1 - a) \log(1 - a).$$

Whenever $H(a) - da^2/2 < 0$, Markov’s inequality rules out independent sets of size an with high probability, so the largest positive root of

$$H(a) - \frac{da^2}{2} = 0$$

serves as a finite- d upper benchmark. At $d = 20$ this root is

$$a_{\text{FM}} = 0.232996,$$

which makes clear that $2\log(20)/20 = 0.2996$ is best read as a large- d asymptotic rather than a tight prediction at moderate d .

3 Computational Design

We compare three methods throughout.

- **Random-order greedy:** 20 random permutations in the notebook run, and 100 in the long-batch runner.
- **Minimum-degree greedy:** repeatedly accept a vertex of current minimum degree and remove it together with its neighbors.
- **IP/CBC:** solve the binary model in PuLP 3.3.0 with CBC 2.10.3, warm-started from the best greedy solution available.

In the independent-set variables, every clique C supplies the valid inequality

$$\sum_{v \in C} x_v \leq 1,$$

since at most one vertex of a clique can be picked. Because we run the vertex-cover model with $y_v = 1 - x_v$, triangle and 4-clique cuts enter in the equivalent form

$$\sum_{v \in C} y_v \geq |C| - 1.$$

We dropped these cuts on the real collaboration network: the graph already carries many coauthorship cliques, and the bare edge formulation was effective enough once the components had been split off.

Every returned set is checked directly against the edge list, and small validation instances are additionally solved by exhaustive search; CBC matched brute force on all three validation graphs,

$$(n, d, \alpha) = (12, 3, 6), \quad (14, 4, 5), \quad (16, 5, 7).$$

All experiments were run on an Apple M3 Pro with 18 GB RAM, Python 3.9.6, PuLP 3.3.0, and CBC 2.10.3. A resumable long-batch script accompanies the notebook for the full grid; it writes one CSV row per instance and picks up without overwriting completed runs.

Table 1: Sparse random graph results for $G(n, 20/n)$. Here W is the warm-start size, L is the IP incumbent, $\Delta = L - W$, and U is the integer upper bound $\lfloor U_{\text{CBC}} \rfloor$ on $\alpha(G)$.

n	edges	rand. mean	rand. best	min-deg	W	L	Δ	status	U	$(\log d/d)n$	FM
100	1023	13.70	17	18	18	20	2	Optimal	20	14.98	23.30
200	2068	29.20	33	37	37	37	0	TimeLimit	59	29.96	46.60
500	5025	75.15	79	95	95	95	0	TimeLimit	166	74.89	116.50
1000	10029	152.35	159	198	198	198	0	TimeLimit	354	149.79	233.00
2000	19862	304.45	313	384	384	384	0	TimeLimit	785	299.57	465.99
3000	30150	454.45	475	584	584	584	0	TimeLimit	1216	449.36	698.99

4 Sparse Random Graph Results

Table 1 reports the completed notebook grid for $d = 20$. The IP time limit was 30 seconds at $n = 100$, 20 seconds for $200 \leq n \leq 2000$, and 60 seconds at $n = 3000$.

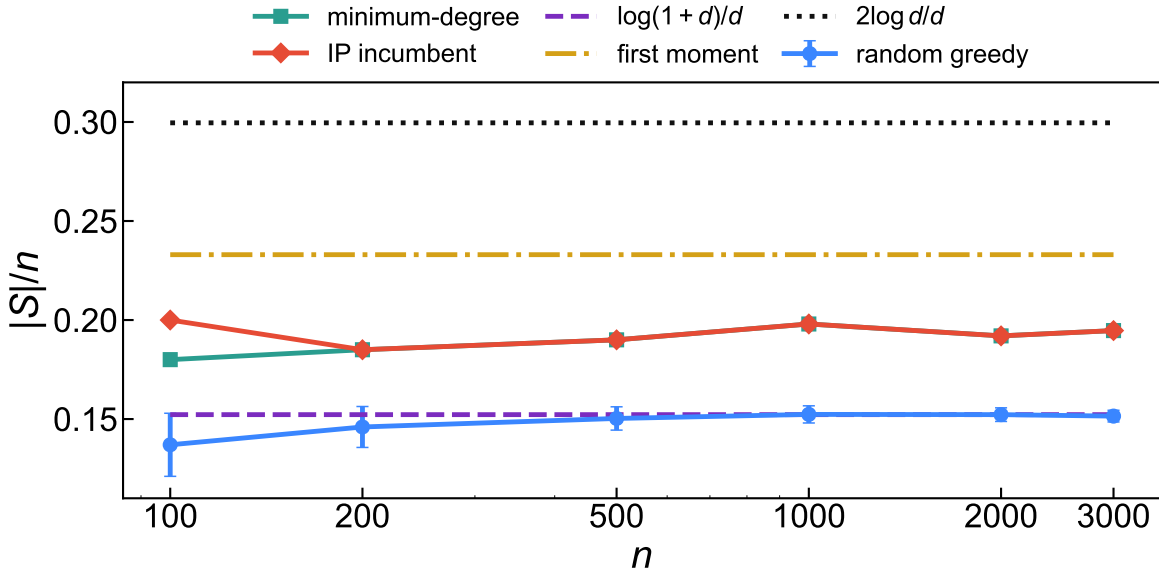


Figure 1: Independent set densities in $G(n, 20/n)$. Random-order greedy tracks the $\log(1+d)/d$ prediction; the IP incumbent follows the stronger minimum-degree warm start and remains below the first-moment benchmark.

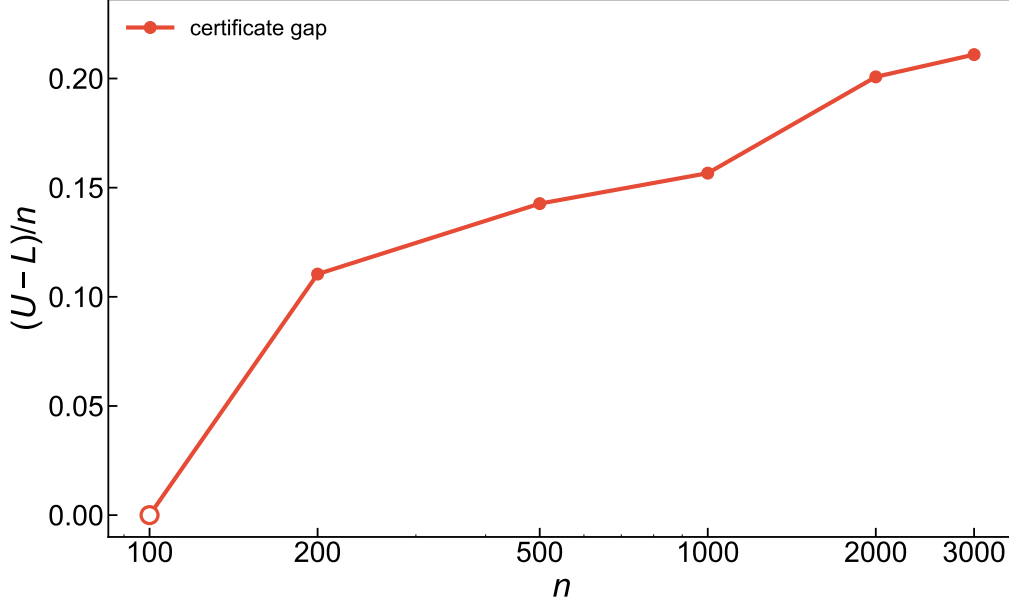


Figure 2: Certificate gap $(U - L)/n$ for the sparse random graph IP runs. The open marker at $n = 100$ denotes a certified optimum; larger instances have incumbents above the greedy threshold but no exact proof within the short time limit.

The random-order greedy mean sits almost exactly on the $\log(1 + d)/d$ scale. At $n = 3000$, for example, the mean is 454.45 against a prediction of $(\log(1 + d)/d)n = 456.68$. Warm-started by minimum-degree greedy, the IP pipeline produces feasible independent sets above the random-order greedy scale on every random instance we tried. In the short-grid runs, though, CBC only manages to improve on the warm start at $n = 100$; for larger n , finding a feasible set is no longer the bottleneck—certifying it is.

5 SNAP ca-GrQc

Our real-network instance is the SNAP ca-GrQc collaboration graph [5], which records coauthorship in the arXiv General Relativity and Quantum Cosmology category; a paper with k authors contributes a k -clique to the graph [6]. After symmetrizing duplicate directed rows, dropping self-loops, and keeping loop-only vertices as isolated nodes, the simple graph has

$$n = 5242, \quad m = 14484, \quad \bar{d} = 5.526.$$

It splits into 355 connected components, the largest of which has 4158 vertices. Since the independence number is additive over components, we solve the IP one component at a time.

The certified value is

$$\alpha(G_{\text{ca-GrQc}}) = 2459,$$

and minimum-degree greedy misses it by a single vertex. Plugging the observed degree $\bar{d}$ into an Erdős–Rényi null model gives

$$\frac{\log \bar{d}}{\bar{d}} n = 1621.6, \quad \frac{\log(1 + \bar{d})}{\bar{d}} n = 1779.4,$$

Table 2: SNAP ca-GrQc independent set results.

quantity	value
random greedy mean, 20 runs	2179.20
random greedy standard deviation	15.17
random greedy best, 20 runs	2202
minimum-degree greedy	2458
IP optimum	2459
IP status	Optimal
total IP time	35.1 seconds

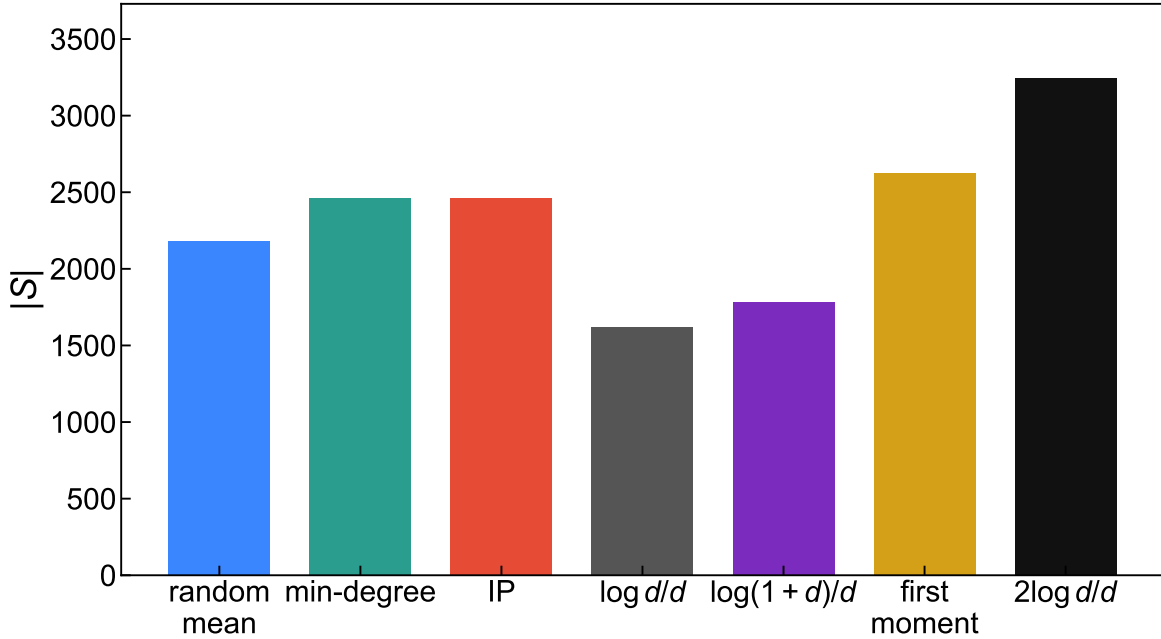


Figure 3: SNAP ca-GrQc compared with random-graph benchmarks using the observed average degree $\bar{d} = 2m/n$.

$$a_{\text{FM}}n = 2625.5, \quad \frac{2 \log \bar{d}}{\bar{d}}n = 3243.2.$$

The true optimum lands between the greedy-scale predictions and the finite- d first-moment benchmark. This is only a null comparison, of course: ca-GrQc has high clustering, many coauthorship cliques, disconnected structure, and a heterogeneous degree distribution, none of which $G(n, d/n)$ captures.

A matched Erdős–Rényi graph with $p = \bar{d}/n$ provides a cleaner null comparison. On this control, CBC times out at 30 seconds with incumbent 2115 and integer upper bound 2605, while the real ca-GrQc instance is solved exactly by component decomposition. The real graph both admits a larger independent set and is easier to certify than the matched ER control, mostly because of its component structure and the geometry typical of collaboration networks.

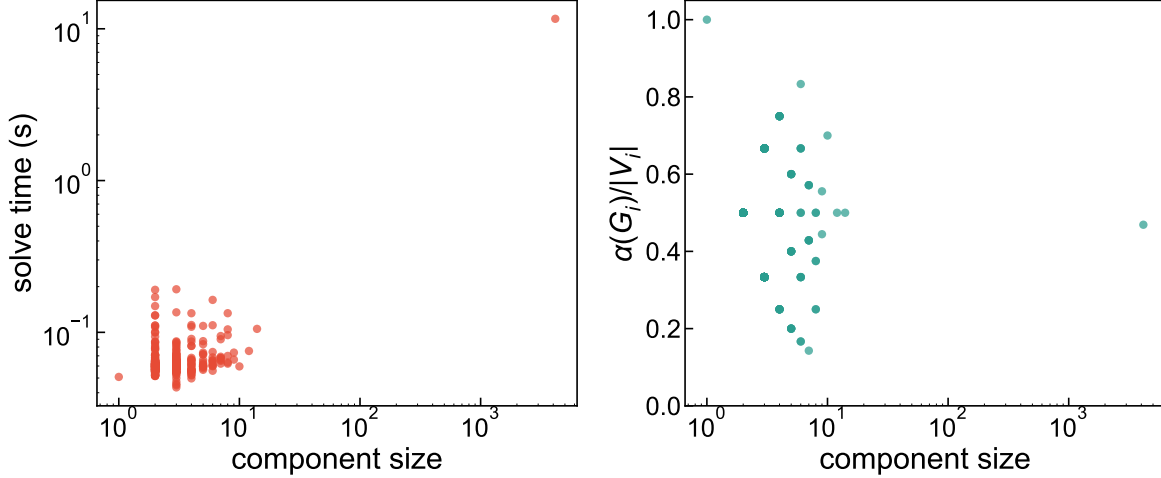


Figure 4: Component-level profile for ca-GrQc. The largest component is the single most expensive subproblem, while many small components contribute small per-component overhead and often have high independent-set density.

Table 3: ca-GrQc and a matched Erdős–Rényi null model generated with the same n and expected average degree $\bar{d}$.

graph	n	m	comps.	rand. mean	min-deg	L	U	status	time
ca-GrQc	5242	14484	355	2179.20	2458	2459	2459	Optimal	35.1s
matched ER	5242	14782	—	1763.15	2115	2115	2605	TimeLimit	32.1s

6 Discussion

The finite- d picture differs visibly from the leading large- d formula. At $d = 20$, the first-moment threshold sits near $0.233n$, well below $2 \log(d)n/d \approx 0.300n$. Our IP incumbents on the short grid lie above random-order greedy but still under this finite- d upper benchmark, which is the qualitative regime one expects.

The bottleneck is certification, not feasibility. The minimum-degree warm start almost always hands the IP a strong feasible solution immediately, but CBC has a much harder time closing the upper-bound side of the certificate. Put in branch-and-bound terms, the primal improves far more quickly than the dual—this is the main algorithmic takeaway from the sparse random instances.

The real-network instance tells a different story. Its 355 components turn what would be a single 5242-node solve into many far smaller subproblems, and the underlying coauthorship structure pushes minimum-degree greedy to within one vertex of optimal. In this regime the random-graph formulas serve as baselines rather than predictions.

7 Conclusion

For $G(n, 20/n)$, the warm-started IP pipeline reliably returns independent sets beyond the random-order greedy scale. Within the completed notebook grid, exact certification only goes through at $n = 100$; the larger instances hit their time limits with nontrivial certificate gaps. On the SNAP ca-GrQc network, the same IP formulation, applied component by component, certifies $\alpha(G) = 2459$.

For broader experiments, the notebook ships with a resumable long-batch runner that supports

multi-seed extensions; the conclusions above are drawn from the completed short-grid, matched-ER, and SNAP runs.

References

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